8 HOMEOSTASIS AND EXCRETION

The kidneys have major roles to play in both homeostasis and excretion. This chapter is mainly concerned with the structure and function of the kidneys. It also deals with another important aspect of homeostasis – maintaining a steady body temperature.

LEARNING OBJECTIVES

- Understand that homeostasis is the maintenance of a constant internal environment
- Understand that control of body water content and body temperature are examples of homeostasis
- Know the excretory products of the lungs, kidneys and skin
- Understand the origin of carbon dioxide and oxygen as waste products of metabolism and their loss from the stomata of a leaf

BIOLOGY ONLY

- Understand that urine contains water, urea and ions
- Understand how the kidney carries out its roles of excretion and osmoregulation
- Describe the structure of the urinary system, including the kidneys, ureters, bladder and urethra

BIOLOGY ONLY

- Describe the structure of a nephron, including the Bowman's capsule and glomerulus, convoluted tubules, loop of Henlé and collecting duct
- Describe ultrafiltration in the Bowman's capsule and the composition of the glomerular filtrate
- Understand why selective reabsorption of glucose occurs at the proximal convoluted tubule
- Understand how water is reabsorbed into the blood from the collecting duct
- Describe the role of ADH in regulating the water content of the blood
- Describe the role of the skin in temperature regulation, with reference to sweating, vasoconstriction and vasodilation

Inside our bodies, conditions are kept relatively constant. This is called **homeostasis**. The kidneys are organs which have a major role to play in both homeostasis and in the removal of waste products, or **excretion**. They filter the blood, removing substances and controlling the concentration of water and solutes (dissolved substances) in the blood and other body fluids.

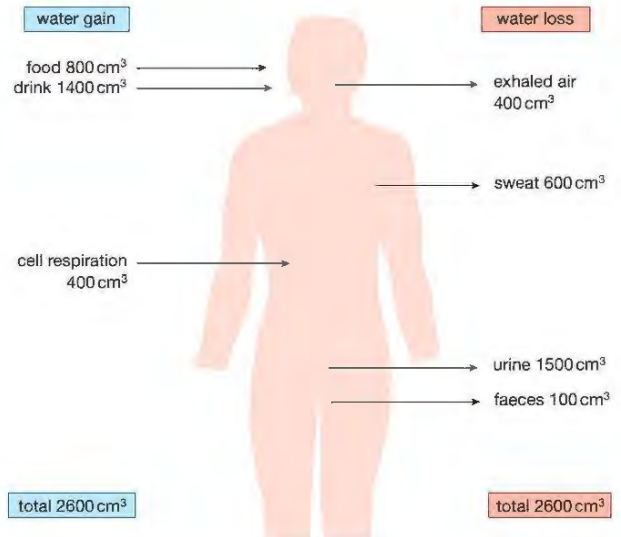
HOMEOSTASIS

KEY POINT

The meaning of 'homeostasis' is 'keeping the conditions in the internal environment of the body relatively constant'. One aspect of homeostasis is the maintenance of the water and salt content of the internal environment. This is called **osmoregulation**.

If you were to drink a litre of water and wait for half an hour, your body would soon respond to this change by producing about the same volume of urine. In other words, it would automatically balance your water input and water loss. Drinking is the main way that our bodies gain water, but there are other sources (Figure 8.1). Some water is present in the food that we eat, and a small amount is formed by cell respiration. The body also loses water, mostly in urine, but also smaller volumes in sweat, faeces and exhaled air. Every day, we gain and lose about the same volume of water, so that the total content of our bodies stays more or less the same. This is an example of homeostasis. The word 'homeostasis' means 'steady state', and refers to keeping conditions inside the body relatively constant.

The inside of the body is known as the **internal environment**. You have probably heard of the 'environment', which means the 'surroundings' of an organism. The *internal* environment is the surroundings of the cells inside the body. It particularly means the blood, together with another liquid called **tissue fluid**.



▲ Figure 8.1 The daily water balance of an adult

DID YOU KNOW?

'Salts' in urine or in the blood are present as ions. For example, the sodium chloride in Table 8.1 will be in solution as sodium ions (Na^+) and chloride ions (Cl^-). Urine contains many other ions, such as potassium (K^+), phosphate (HPO_4^{2-}) and ammonium (NH_4^+), and removes excess ions from the blood.

Tissue fluid is a watery solution of salts, glucose and other solutes. It surrounds all the cells of the body, forming a pathway for the transfer of nutrients between the blood and the cells. Tissue fluid is formed by leakage from blood capillaries. It is similar in composition to blood plasma, but lacks the plasma proteins.

It is not just water and salts that are kept constant in the body. Many other components of the internal environment are maintained. For example, the level of carbon dioxide in the blood is regulated, along with the blood pH, the concentration of dissolved glucose (see Chapter 7) and the body temperature.

Homeostasis is important because cells will only function properly if they are bathed in a tissue fluid which provides them with their optimum conditions. For instance, if the tissue fluid contains too many solutes, the cells will lose water by osmosis, and become dehydrated. If the tissue fluid is too dilute, the cells will swell up with water. Both conditions will prevent them working efficiently and might cause permanent damage. If the pH of the tissue fluid is not correct, it will affect the activity of the cell's enzymes, as will a body temperature much different from 37 °C. It is also important that excretory products are removed. Substances such as urea must be prevented from building up in the blood and tissue fluid, where they would be toxic to cells.

URINE

An adult human produces about 1.5 dm^3 of urine every day, although this volume depends very much on the amount of water drunk and the volume lost in other forms, such as sweat. Every litre of urine contains about 40 g of waste products and salts (Table 8.1).

KEY POINT

If you are studying for the Double Award, you only need to be aware that urea is an excretory product of the kidneys.

Table 8.1 Some of the main solutes in urine.

Substance	Amount / g per dm^3
urea	23.3
ammonia	0.4
other nitrogenous waste	1.6
sodium chloride (salt)	10.0
potassium	1.3
phosphate	2.3

Notice the words *nitrogenous waste*. Urea and ammonia are two examples of nitrogenous waste. It means that they contain the element nitrogen. All animals have to excrete a nitrogenous waste product.

KEY POINT

Excretion is the process by which waste products of metabolism are removed from the body. In humans, the main nitrogenous excretory substance is **urea**. Another excretory product is carbon dioxide, produced by cell respiration and excreted by the lungs (Chapter 3). The human skin is also an excretory organ, since the sweat that it secretes contains small amounts of urea.

Plants also excrete waste products of metabolism. In light, their leaves produce oxygen from photosynthesis. In the dark they excrete carbon dioxide from respiration. Plants exchange these gases through pores in the leaf called stomata (see Chapter 11).

The reason behind this is quite complicated. Carbohydrates and fats only contain the elements carbon, hydrogen and oxygen. However, proteins also contain nitrogen. If the body has too much carbohydrate or fat, these substances can be stored, for example as glycogen in the liver, or as fat under the skin and around other organs. Excess proteins, or their building blocks (called amino acids) *cannot* be stored. The amino acids are first broken down in the liver. They are converted into carbohydrate (which is stored as glycogen) and the main nitrogen-containing waste product, urea. The urea passes into the blood, to be filtered out by the kidneys during the formation of urine. Notice that the urea is made by chemical reactions in the cells of the body (the body's metabolism). 'Excretion' means getting rid of waste of this kind. When the body gets rid of solid waste from the digestive system (faeces), this is not excretion, since it contains few products of *metabolism*, just the 'remains' of undigested food, along with bacteria and dead cells.

So the kidney is in fact carrying out two functions. It is a *homeostatic* organ, controlling the water and salt (ion) concentration in the body as well as an *excretory* organ, concentrating nitrogenous waste in a form that can be eliminated.

BIOLOGY ONLY

THE URINARY SYSTEM

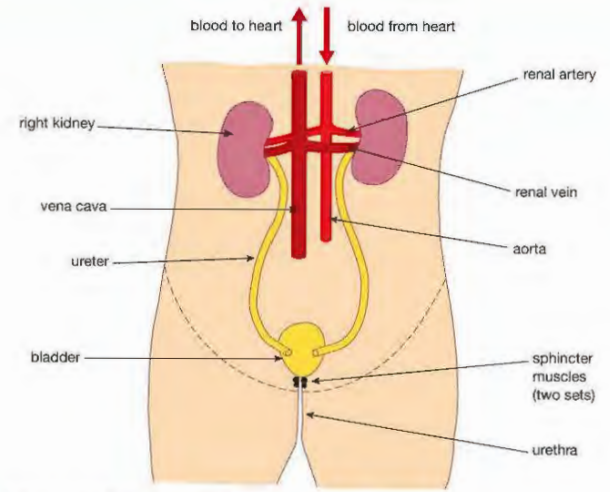
The human urinary system is shown in Figure 8.2.

Each kidney is supplied with blood through a short **renal artery**. This leads straight from the body's main artery, the aorta, so the blood entering the kidney is at a high pressure. Inside each kidney the blood is filtered, and the 'cleaned' blood passes out through each **renal vein** to the main vein, or vena cava. The urine leaves the kidneys through two tubes, the **ureters**, and is stored in a muscular bag called the **bladder**.

The bladder has a tube leading to the outside, called the **urethra**. The wall of the urethra contains two ring-shaped muscles, called **sphincter muscles**. They can contract to close the urethra and hold back the urine. The lower sphincter muscle is under conscious control, or 'voluntary', while the upper one is involuntary – it automatically relaxes when the bladder is full.

DID YOU KNOW?

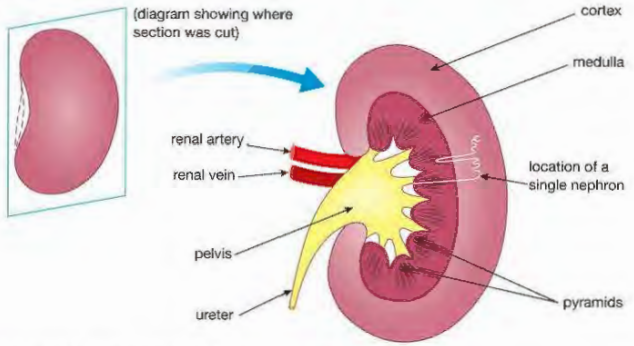
A baby cannot control its voluntary sphincter. When the bladder is full, the baby's involuntary sphincter relaxes, releasing the urine. A toddler learns to control this muscle and hold back the urine.



▲ Figure 8.2 The human urinary system

THE KIDNEYS

If you were to cut a kidney lengthwise you would be able to see the structures shown in Figure 8.3.

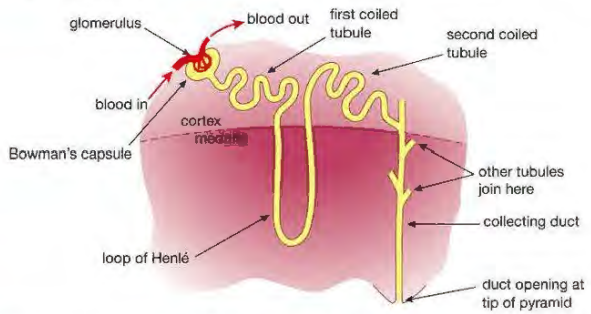


▲ Figure 8.3 Section through a kidney cut along the plane shown.

There is not much that you can make out without the help of a microscope. The darker outer region is called the **cortex**. This contains many tiny blood vessels that branch from the renal artery. It also contains microscopic tubes that are not blood vessels. They are the filtering units, called kidney tubules or **nephrons** (from the Greek word *nephros*, meaning kidney). The tubules then run down through the middle layer of the kidney, called the **medulla**. The medulla has bulges called 'pyramids' pointing inwards towards the concave side of the kidney. The tubules in the medulla eventually join up and lead to the tips of these pyramids, where they empty urine into a funnel-like structure called the **pelvis**. The pelvis connects with the ureter, carrying the urine to the bladder.

STRUCTURE OF THE NEPHRON

By careful dissection, biologists have been able to work out the structure of a single tubule and its blood supply (Figure 8.4). There are about a million of these in each kidney.



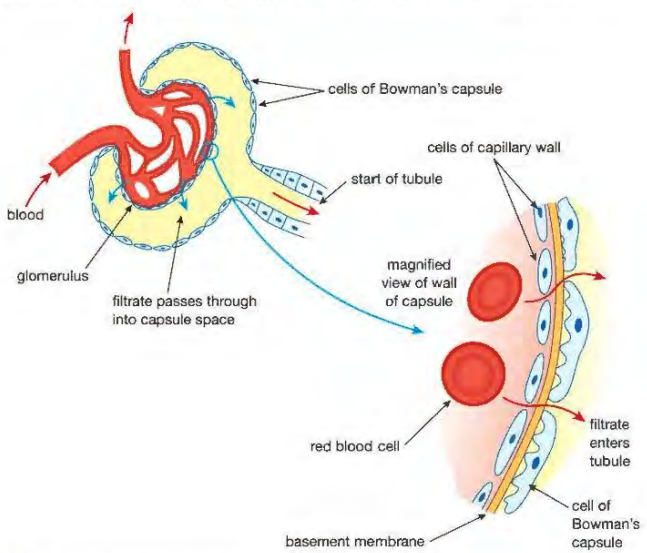
▲ Figure 8.4 A single nephron, showing its position in the kidney. Each kidney contains about a million of these filtering units.

EXTENSION WORK

The first and second coiled tubules are also known as the proximal and distal convoluted tubules. 'Proximal' means near the start of the tubule, and 'distal' means near the end.

ULTRAFILTRATION IN THE BOWMAN'S CAPSULE

At the start of the nephron is a hollow cup of cells called the **Bowman's capsule**. It surrounds a ball of blood capillaries called a **glomerulus** (plural glomeruli). It is here that the blood is filtered. Blood enters the kidney through the renal artery, which divides into smaller and smaller arteries. The smallest arteries (arterioles) supply the capillaries of the glomerulus (Figure 8.5).



▲ Figure 8.5 A Bowman's capsule and glomerulus

EXTENSION WORK

The cells of the glomerulus capillaries do not fit together very tightly, there are spaces between them making the capillary walls much more permeable than others in the body. The cells of the Bowman's capsule also have gaps between them, so only act as a coarse filter. It is the basement membrane that acts as the fine molecular filter.

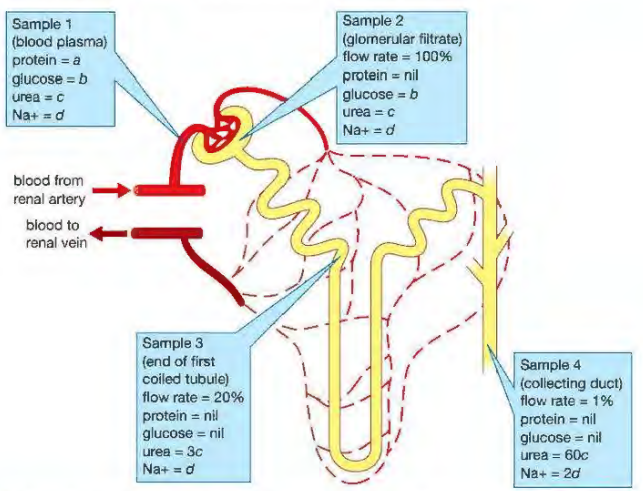
A blood vessel with a smaller diameter carries blood away from the glomerulus, leading to capillary networks which surround the other parts of the nephron. Because of the resistance to flow caused by the glomerulus, the pressure of the blood in the arteriole leading to the glomerulus is very high. This pressure forces fluid from the blood through the walls of the capillaries and the Bowman's capsule, into the space in the middle of the capsule. Blood in the glomerulus and the space in the capsule are separated by two layers of cells, the capillary wall and the wall of the capsule. Between the two cell layers is a third layer called the **basement membrane**, which is not made of cells. These layers act like a filter, allowing water, ions, and small molecules like glucose and urea to pass through, but holding back blood cells and large molecules such as proteins. The fluid that enters the capsule space is called the **glomerular filtrate**. This process, where the filter separates different-sized molecules under pressure, is called **ultrafiltration**.

CHANGES TO THE FILTRATE IN THE REST OF THE NEPHRON

The kidneys produce about 125 cm^3 (0.125 dm^3) of glomerular filtrate per minute. This works out at 180 dm^3 per day. Remember though, only 1.5 dm^3 of urine is lost from the body every day, which is less than 1% of the volume filtered through the capsules. The other 99% of the glomerular filtrate is *reabsorbed* back into the blood.

We know this because scientists have actually analysed samples of fluid from the space in the middle of the nephron. Despite the diameter of the space being only 20 μm (0.02 mm), it is possible to pierce the tubule with microscopic glass pipettes and extract the fluid for analysis. Figure 8.6 shows the structure of the nephron and the surrounding blood vessels in more detail.

There are two coiled regions of the tubule in the cortex, separated by a U-shaped loop that runs down into the medulla of the kidney, called the **loop of Henlé**. After the second coiled tubule, several nephrons join up to form a **collecting duct**, where the final urine passes out into the pelvis.



▲ Figure 8.6 A nephron and its blood supply. Samples 1–4 show what is happening to the fluid as it travels along the nephron.

Samples 1–4 show the results of analysing the blood before it enters the glomerulus, and the fluid at three points inside the tubule. The flow rate is a measure of how much water is in the tubule. If the flow rate falls from 100% to 50%, this is because 50% of the water in the tubule has passed back into the blood. To make the explanation easier, the concentrations of dissolved protein, glucose, urea and sodium are shown by different letters (*a* to *d*). You can tell the relative concentration of one substance at different points along the tubule from this. For example, urea at a concentration '3*c*' is three times more concentrated than when it is '*c*'.

In the blood (sample 1) the plasma contains many dissolved solutes, including protein, glucose, urea and salts (just sodium ions, Na^+, are shown here). As we saw above, protein molecules are too big to pass through into the tubule, so the protein concentration in sample 2 is zero. The other substances are at the same concentration as in the blood.

KEY POINT

The kidney tubule reabsorbs different amounts of various substances. This is called **selective reabsorption**. Sodium ions and glucose are reabsorbed by active transport.

Now look at sample 3, taken at the end of the first coiled part of the tubule. The flow rate that was 100% is now 20%. This must mean that 80% of the water in the tubule has been reabsorbed back into the blood. If no solutes were reabsorbed along with the water, their concentrations should be *five times* what they were in sample 2. Since the concentration of sodium hasn't changed, 80% of this substance must have been reabsorbed (and some of the urea too). However, the glucose concentration is now zero – *all* of the glucose is taken back into the blood in the first coiled tubule. This is necessary because glucose is a useful substance that is needed by the body.

Finally, look at sample 4. By the time the fluid passes through the collecting duct, its flow rate is only 1%. This is because 99% of the water has been reabsorbed. Protein and glucose are still zero, but most of the urea is still in the fluid. The level of sodium is only 2*d*, so not all of it has been reabsorbed, but it is still twice as concentrated as in the blood.

This description has only looked at a few of the more important substances. Other solutes are concentrated in the urine by different amounts. Some, like ammonium ions, are secreted *into* the fluid as it passes along the tubule. The concentration of ammonium ions in the urine is about 150 times what it is in the blood.

THE LOOP OF HENLÉ

You might be wondering what the role of the loop of Henlé is. The full answer to this is rather complicated. You may meet it again if you study biology beyond International GCSE, but for now a brief explanation will be enough. It is involved with concentrating the fluid in the tubule by causing more water to be reabsorbed into the blood. Mammals with long loops of Henlé can make a more concentrated urine than ones with short loops. Desert animals have many long loops of Henlé, so they are able to produce very concentrated urine, conserving water in their bodies. Animals which have easy access to water, such as otters or beavers, have short loops of Henlé. Humans have a mixture of long and short loops.

KEY POINT

Here is a summary of what happens in the kidney nephron.

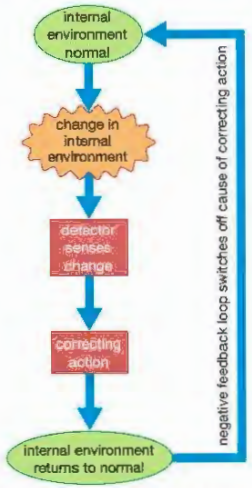
Part of the plasma leaves the blood in the Bowman's capsule and enters the nephron. The filtrate consists of water and small molecules. As the fluid passes along the nephron, all the glucose is absorbed back into the blood in the first coiled part of the tubule, along with most of the sodium and chloride ions. In the rest of the tubule,

KEY POINT

Maintaining a constant water and salt content in the body is known as osmoregulation. 'Regulation' means that they are kept at a certain level.

EXTENSION WORK

As well as causing the pituitary gland to release ADH, the receptor cells in the hypothalamus also stimulate a 'thirst centre' in the brain. This makes the person feel thirsty, so that they will drink water, diluting the blood.



▲ Figure 8.7 In homeostasis, the extent of a correction is monitored by negative feedback.

CONTROL OF THE BODY'S WATER CONTENT

Not only can the kidney produce urine that is more concentrated than the blood, it can also *control* the concentration of the urine, and so *regulate* the water content of the blood. This chapter began by asking you to think what would happen if you drank a litre of water. The kidneys respond to this 'upset' to the body's water balance by making a larger volume of more dilute urine. Conversely, if the blood becomes too concentrated, the kidneys produce a smaller volume of urine. These changes are controlled by a hormone produced by the pituitary gland, at the base of the brain. The hormone is called **antidiuretic hormone**, or **ADH**.

'Diuresis' means the flow of urine from the body, so 'antidiuresis' means producing less urine. ADH starts to work when your body loses too much water, for example if you are sweating heavily and not replacing lost water by drinking.

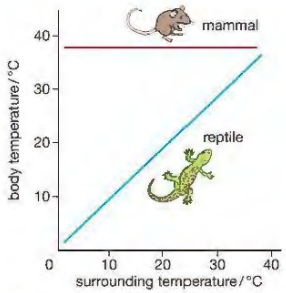
The loss of water means that the concentration of the blood starts to increase. This is detected by receptor cells in a region of the brain called the hypothalamus, situated above the pituitary gland. These cells are sensitive to the solute concentration of the blood, and cause the pituitary gland to release more ADH. The ADH travels in the bloodstream to the kidney. At the kidney tubules it causes the collecting ducts to become more permeable to water, so that more water is reabsorbed back into the blood. This makes the urine more concentrated, so that the body loses less water and the blood becomes more dilute.

The action of ADH illustrates the principle of **negative feedback**. A change in conditions in the body is detected, and starts a process that works to return conditions to normal. When the conditions are returned to normal, the corrective process is switched off (Figure 8.7).

In the situation described in the text, where the body loses too much water, the blood becomes too concentrated. This switches on ADH release, which acts at the kidneys to correct the problem. The word 'negative' means that the process works to eliminate the change. When the blood returns to normal, ADH release is switched off. The feedback pathway forms a 'closed loop'. Many conditions in the body are regulated by negative feedback loops like this.

When the water content of the blood returns to normal, this acts as a signal to 'switch off' the release of ADH. The kidney tubules then reabsorb less water. Similarly, if someone drinks a large volume of water, the blood will become too dilute. This leads to lower levels of ADH secretion, the kidney tubules become less permeable to water, and more water passes out of the body in the urine. In this way, through the action of ADH, the level of water in the internal environment is kept constant.

END OF BIOLOGY ONLY



▲ Figure 8.8 The temperature of a homeotherm, such as a mammal, is kept constant at different external temperatures, whereas the lizard's body temperature changes.

DID YOU KNOW?

Physiology is the branch of biology that deals with how the bodies of animals or plants work, for example how muscles contract, how nerves send impulses, or how xylem carries water through plants. In this chapter you have read about kidney physiology.

CONTROL OF BODY TEMPERATURE

You may have heard mammals and birds described as 'warm blooded'. A better word for this is **homeothermic**. It means that they keep their body temperature constant, despite changes in the temperature of their surroundings. For example, the body temperature of humans is kept steady at about 37 °C, give or take a few tenths of a degree. This is another example of homeostasis. All other animals are 'cold blooded'. For example, if a lizard is kept in an aquarium at 20 °C, its body temperature will be 20 °C too. If the temperature of the aquarium is raised to 25 °C, the lizard's body temperature will rise to 25 °C as well. We can show this difference between **homeotherms** and other animals as a graph (Figure 8.8).

In the wild, lizards keep their temperature more constant than in Figure 8.8, by adapting their behaviour. For example, in the morning they may stay in the sun to warm their bodies, or at midday, if the sun is too hot, retreat to holes in the ground to cool down.

The real difference between homeotherms and all other animals is that homeotherms can keep their temperatures constant by using *physiological* changes for generating or losing heat. For this reason, mammals and birds are also called 'endotherms', meaning 'heat from inside'.

An endotherm uses heat from the chemical reactions in its cells to warm its body. It then controls its heat loss by regulating processes like sweating and blood flow through the skin. Endotherms use behavioural ways to control their temperature too. For example, penguins 'huddle' together in groups to keep warm, and humans put on extra clothes in winter.

What is the advantage of a human maintaining a body temperature of 37 °C? It means that all the chemical reactions taking place in the cells of the body can go on at a steady, predictable rate. The metabolism doesn't slow down in cold environments. If you watch goldfish in a garden pond, you will notice that in summer, when the pond water is warm, they are very active, swimming about quickly. In winter, when the temperature drops, the fish slow down and become very sluggish in their actions. This would happen to a mammal too, if its body temperature varied.

It is also important that the body does not become *too* hot. The cells' enzymes work best at 37 °C. At higher temperatures enzymes, like all proteins, are destroyed by **denaturing** (see Chapter 1). Endotherms have all evolved a body temperature of around 40 °C (Table 8.2) and enzymes that work best at this temperature.

Table 8.2 The body temperatures of a range of mammals and birds

Species	Average and normal range of body temperature / °C	Species	Average and normal range of body temperature / °C
brown bear	38.0 ± 1.0	shrew	35.7 ± 1.2
camel	37.5 ± 0.5	whale	35.7 ± 0.1
elephant	36.2 ± 0.5	duck	43.1 ± 0.3
fox	38.8 ± 1.3	ostrich	39.2 ± 0.7
human	36.9 ± 0.7	penguin	39.0 ± 0.2
mouse	39.3 ± 1.3	thrush	40.0 ± 1.7
polar bear	37.5 ± 0.4	wren	41.0 ± 1.0

MONITORING BODY TEMPERATURE

In humans and other mammals the core body temperature is monitored by a part of the brain called the **thermoregulatory centre**. This is located in the hypothalamus of the brain. It acts as the body's 'thermostat'.

If a person goes into a warm or cold environment, the first thing that happens is that temperature receptors in the skin send electrical impulses to the hypothalamus, which stimulates the brain to alter our behaviour. We start to feel hot or cold, and usually do something about it, such as finding shade, having a cold drink, or putting on more clothes.

If changes to our behaviour are not enough to keep our body temperature constant, the thermoregulatory centre in the hypothalamus detects a change in the temperature of the blood flowing through it. It then sends signals via nerves to other organs of the body, which regulate the temperature by physiological means.

DID YOU KNOW?

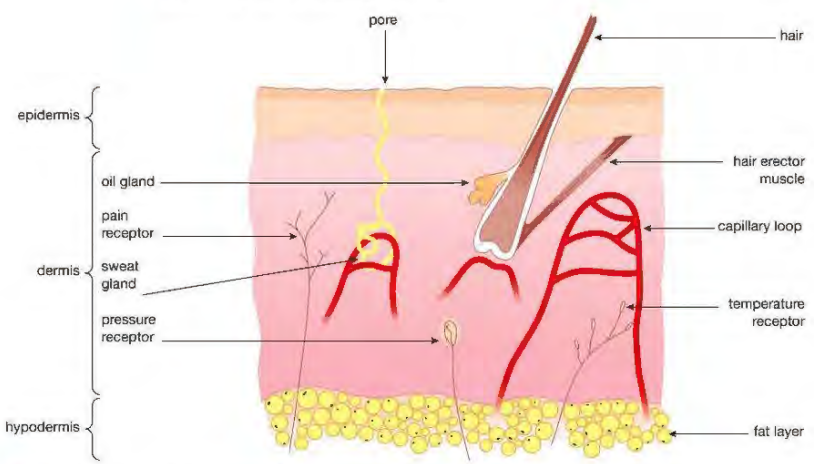
A thermostat is a switch that is turned on or off by a change in temperature. It is used in electrical appliances to keep their temperature steady. For example, a thermostat in an iron can be set to 'hot' or 'cool' to keep the temperature of the iron set for ironing different materials.

THE SKIN AND TEMPERATURE CONTROL

The human skin has a number of functions related to the fact that it forms the outer surface of the body. These include:

- forming a tough outer layer able to resist mechanical damage
- acting as a barrier to the entry of pathogens
- forming an impermeable surface, preventing loss of water
- acting as a sense organ for touch and temperature changes
- controlling the loss of heat through the body surface.

Figure 8.9 shows the structure of human skin. It is made up of three layers: the epidermis, dermis and hypodermis.



▲ Figure 8.9 A section through human skin

The outer **epidermis** consists of dead cells that stop water loss and protect the body against invasion by microorganisms such as bacteria. The **hypodermis** contains fatty tissue, which insulates the body against heat loss and is a store of energy. The middle layer, the **dermis**, contains many sensory receptors. It is also the location of sweat glands and many small blood vessels, as well as hair follicles. These last three structures are involved in temperature control.

Imagine that the hypothalamus detects a rise in the central (core) body temperature. Immediately it sends nerve impulses to the skin. These bring about changes to correct the rise in temperature.

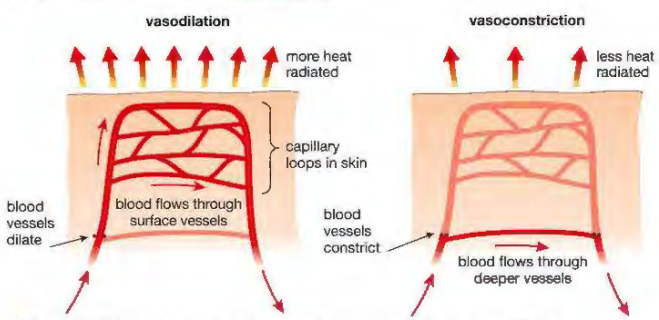
First of all, the **sweat glands** produce greater amounts of sweat. This liquid is secreted onto the surface of the skin. When a liquid evaporates, it turns into a gas. This change needs energy, called the latent heat of vaporisation. When sweat evaporates, the energy is supplied by the body's heat, cooling the body down. It is not that the sweat is cool – it is secreted at body temperature. It only has a cooling action when it evaporates. In very humid atmospheres (e.g. a tropical rainforest) the sweat stays on the skin and doesn't evaporate. It then has very little cooling effect.

Secondly, hairs on the surface of the skin lie flat against the skin's surface. This happens because of the relaxation of tiny muscles called **hair erector muscles** attached to the base of each hair. In cold conditions, these contract and the hairs are pulled upright. The hairs trap a layer of air next to the skin, and since air is a poor conductor of heat, this acts as insulation. In warm conditions, the thinner layer of trapped air means that more heat will be lost. This is not very effective in humans, because the hairs over most of our body do not grow very large. It is very effective in hairy mammals like cats or dogs. The same principle is used by birds, which 'fluff out' their feathers in cold weather.

HINT

Students often describe vasodilation incorrectly. They talk about the blood vessels 'moving nearer the surface of the skin'. They don't *move* at all, it's just that more blood flows through the surface vessels.

Lastly, there are tiny blood vessels called capillary loops in the dermis. Blood flows through these loops, radiating heat to the outside, and cooling the body down. If the body is too hot, arterioles (small arteries) leading to the capillary loops dilate (widen). This increases the blood flow to the skin's surface (Figure 8.10) and is called **vasodilation**.



▲ Figure 8.10 Blood flow through the surface of the skin is controlled by vasodilation or vasoconstriction.

In cold conditions, the opposite happens. The arterioles leading to the surface capillary loops constrict (become narrower) and blood flow to the surface of the skin is reduced, so that less heat is lost. This is called **vasoconstriction**. Vasoconstriction and vasodilation are brought about by tiny rings of muscles in the walls of the arterioles, called sphincter muscles, like the sphincters you read about earlier in this chapter, at the outlet of the bladder.

There are other ways that the body can control heat loss and heat gain. In cold conditions, the body's metabolism speeds up, generating more heat. The liver, a large organ, can produce a lot of metabolic heat in this way. The hormone adrenaline stimulates the increase in metabolism (see Chapter 7). **Shivering** also takes place, where the muscles contract and relax rapidly. This also generates a large amount of heat.

Sweating, vasodilation and vasoconstriction, hair erection, shivering and changes to the metabolism, along with behavioural actions, work together to keep the body temperature to within a few tenths of a degree of the 'normal' 37 °C. If the difference is any bigger than this it shows that something is wrong. For instance, a temperature of 39 °C might be due to an illness.

CHAPTER QUESTIONS

More questions on homeostasis and excretion can be found at the end of Unit 2 on page 130.

SKILLS CRITICAL THINKING

1 Which of the following is *not* an example of excretion?

A loss of carbon dioxide by a plant in the dark
B removal of carbon dioxide from the lungs
C removal of urea in urine
D elimination of faeces from the alimentary canal

BIOLOGY ONLY

2 The following structures are parts of the human urinary system.

1. ureter
2. kidney
3. urethra
4. bladder

In which order does a molecule of urea pass through these structures?

A 2 → 1 → 4 → 3
B 2 → 3 → 4 → 1
C 1 → 3 → 2 → 4
D 3 → 4 → 1 → 2

3 Below are three statements about the action of antidiuretic hormone (ADH).

1. More ADH is released when the water content of the blood rises
2. ADH increases the permeability of the collecting duct
3. When ADH is released, more water is reabsorbed.

Which of these statements are true?

A 1 and 2
B 1 and 3
C 2 and 3
D 1, 2 and 3

END OF BIOLOGY ONLY

SKILLS CRITICAL THINKING (7)

4 If the human body temperature starts to rise, which of the following happens?

A vasoconstriction of arterioles in the skin
B contraction of hair erector muscles
C decrease in the rate of metabolism
D decrease in production of sweat by glands in the skin

BIOLOGY ONLY

5 Explain the meaning of the following terms:

a homeostasis (5)

b excretion (6)

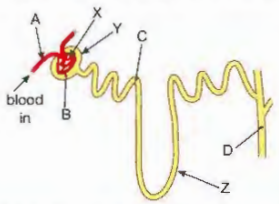
c ultrafiltration (5)

d selective reabsorption (7)

e endotherm.

SKILLS ANALYSIS (5)

6 The diagram below shows a simple diagram of a nephron (kidney tubule).



a What are the names of the parts labelled X, Y and Z?

(6) b Four places in the nephron and its blood supply are labelled A, B, C and D. Which of the following substances are found at each of these four places?

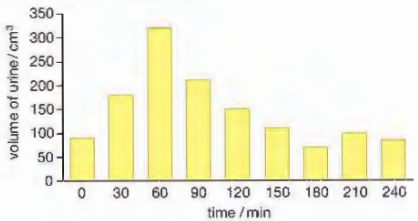
water urea protein glucose salt

SKILLS CRITICAL THINKING, REASONING (10)

7 The hormone ADH controls the amount of water removed from the blood by the kidneys. Write a short description of the action of ADH in a person who has lost a lot of water by sweating, but has been unable to replace this water by drinking. Explain how this is an example of negative feedback. (You will need to write about 250 words to answer this question fully.)

SKILLS ANALYSIS

8 The bar chart shows the volume of urine collected from a person before and after drinking 1000 cm^3 (1 dm^3) of distilled water. The person's urine was collected immediately before the water was drunk and then at 30-minute intervals for four hours.



SKILLS ANALYSIS (5)

a Describe how the output of urine changed during the course of the experiment.

(9) b Explain the difference in urine produced at 60 minutes and at 90 minutes.

SKILLS REASONING

c The same experiment was repeated with the person sitting in a very hot room. How would you expect the volume of urine collected to differ from the first experiment? Explain your answer.

SKILLS PROBLEM SOLVING

d Between 90 and 120 minutes, the person produced 150 cm^3 of urine. If the rate of filtration at the glomeruli during this time was 125 cm^3 per minute, calculate the percentage of filtrate reabsorbed by the kidney tubules.

END OF BIOLOGY ONLY

SKILLS INTERPRETATION (7)

9 Construct a table like the one below to show the changes that take place when a person is put in a hot or cold environment. Your table should have three columns.

Changes taking place	Hot environment	Cold environment
sweating		
blood flow through capillary loops		vasoconstriction decreases blood flow through surface capillaries so that less heat is radiated from the skin
hairs in skin		
shivering		
metabolism		

SKILLS ANALYSIS, REASONING (5)

10 Look at the body temperatures of mammals and birds shown in Table 8.2 on page 112. Use the information in the table to answer these questions:

a How does the average temperature of birds differ from the average temperature of mammals? Can you suggest why this is an advantage for birds?

b Is there a relationship between the body temperature of a mammal and the temperature of its habitat? Give an example to support your answer.

(6) c Polar bears have thick white fur covering their bodies. Explain two ways in which this is an adaptation to their habitat (the place where the animal lives).